

# EMPLOYEE MANAGEMENT SYSTEM USING SQL

## Project Summary – MySQL Assignment 2: Querying Data

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### 1. Project Overview

This project focuses on designing and querying a relational database using MySQL. It demonstrates fundamental and advanced SQL concepts by building an Employee Management System and performing various data operations.

The system consists of three main tables:

- Departments – Stores department details
  - Locations – Stores office location data
  - Employees – Stores employee information such as salary, designation, and relationships with departments and locations
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### 2. Objective

The primary objectives of this project are:

- To understand and apply SQL query techniques
  - To perform data filtering, sorting, and aggregation
  - To work with relationships using joins
  - To implement advanced queries using window functions
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### 3. Tools & Technologies

- Database: MySQL

- Language: SQL
  - Platform: MySQL Workbench / Command Line
  - Documentation: MS Word
- 

## **4. Database Design**

- Created structured tables with Primary Keys and Foreign Keys
  - Applied constraints such as:
    - NOT NULL
    - UNIQUE
    - CHECK
  - Maintained data integrity and relationships between tables
- 

## **5. Data Operations**

- Inserted multiple employee records with various roles and salaries
  - Managed NULL values (e.g., missing designation)
  - Updated and maintained structured datasets
- 

## **6. SQL Query Implementation**

### **6.1 Clauses & Operators**

- Retrieved distinct salary values using DISTINCT
- Used column aliases (AS) for better readability
- Applied filtering using WHERE conditions
  - Example: salary > 50000
  - Hire date before 2016-01-01

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## 6.2 Sorting & Limiting

- Sorted employees using ORDER BY
  - Department (ascending)
  - Salary (descending)
- Limited results using LIMIT
  - Example: Top 5 employees hired in 2018

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## 6.3 Aggregate Functions

- Calculated total salary using SUM()
- Found minimum age using MIN()
- Calculated average salary using AVG()
- Used GROUP BY for categorization

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## 6.4 HAVING Clause

- Filtered grouped results
  - Departments with fewer than 3 employees
  - Locations where average age < 30

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## 6.5 Joins

- **INNER JOIN**
  - Combined employee and department data
- **LEFT JOIN**
  - Displayed all departments including those without employees

- **RIGHT JOIN**
    - Displayed all locations including those without employees
  - **CROSS JOIN**
    - Generated all possible combinations of departments and locations
  - **SELF JOIN**
    - Compared employees within the same department
- 

## 6.6 Window Functions

- Used RANK()
    - Ranked employees based on salary
  - Used DENSE\_RANK()
    - Ranked employees within each department
  - Running total calculation
    - Calculated cumulative salary by department
- 

## 7. Tasks & Screenshots

The project includes execution screenshots for:

- DISTINCT queries
- Alias usage
- WHERE conditions
- ORDER BY & LIMIT
- Aggregate functions
- GROUP BY & HAVING
- Joins (all types)

- Window functions

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## CLAUSE & OPERATORS

### 1. DISTINCT VALUES:

- A query to retrieve distinct salaries from the Employees table.

#### SCREENSHOT:

The screenshot shows a SQL query editor with the following code:

```
89  /* Clause & Operators: */
90
91  /*Distinct Clause*/
92
93  • select distinct salary from employees;
94
95
```

Below the editor is a toolbar with options: Result Grid, Filter Rows, Export, and Wrap Cell Content. The results are displayed in a table with the following data:

| salary    |
|-----------|
| 60000.00  |
| 80000.00  |
| 45000.00  |
| 70000.00  |
| 55000.00  |
| 65000.00  |
| 90000.00  |
| 95000.00  |
| 75000.00  |
| 85000.00  |
| 100000.00 |

### 2. ALIAS (AS):

- Provide aliases for the "age" and "salary" columns as "Employee\_Age" and "Employee\_Salary", respectively.

#### SCREENSHOT OF EMPLOYEE\_AGE:

Query 1 Employee Database S2\* x

Limit to 1000 rows

```
96 /*Distinct Clause*/
97
98 • select distinct salary from employees;
99
100 /*Alias (as)*/
101
102 • select age as Employee_age from employees;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: I A

|   | Employee_age |
|---|--------------|
| ▶ | 27           |
|   | 31           |
|   | 26           |
|   | 29           |
|   | 25           |
|   | 28           |
|   | 32           |
|   | 27           |
|   | 30           |
|   | 29           |
|   | 28           |
|   | 26           |
|   | 33           |
|   | 30           |
|   | 27           |
|   | 31           |
|   | 28           |
|   | 25           |
|   | 22           |

employees 5 x

**SCREENSHOT OF EMPLOYEE\_SALARY:**

Query 1 Employee Database SZ x

Limit to 1000 rows

```

99
100  /*Alias (as)*/
101
102 •  select age as Employee_age from employees;
103
104 •  select salary as Employee_salary from employees;
105

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: [IA](#)

|   | Employee_salary |
|---|-----------------|
| ▶ | 60000.00        |
|   | 80000.00        |
|   | 45000.00        |
|   | 70000.00        |
|   | 55000.00        |
|   | 65000.00        |
|   | 90000.00        |
|   | 70000.00        |
|   | 55000.00        |
|   | 80000.00        |
|   | 60000.00        |
|   | 95000.00        |
|   | 75000.00        |
|   | 60000.00        |
|   | 85000.00        |
|   | 100000.00       |
|   | 80000.00        |
|   | 75000.00        |
|   | 55000.00        |

### 3. WHERE CLAUSE & OPERATORS:

- Retrieve employees with a salary greater than ₹50000 and hired before 2016-01-01.

**SCREENSHOT:**

Employee Database S2\*

Limit to 1000 rows

```

101
102 • select age as Employee_age from employees;
103
104                               /*WHERE CLAUSE & OPERATORS: */
105 -- where & and operator--
106
107 • select * from employees where salary >50000 and hire_date <'2016-01-01';

```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: IA

|   | employee_id | employee_name  | gender | age  | hire_date  | designation                    | salary   | department_id | location_id |
|---|-------------|----------------|--------|------|------------|--------------------------------|----------|---------------|-------------|
| ▶ | 5001        | Vihaan Singh   | M      | 27   | 2015-01-20 | Data Analyst                   | 60000.00 | 3             | 4           |
|   | 5002        | Reyansh Singh  | M      | 31   | 2015-03-10 | Network Engineer               | 80000.00 | 12            | 1           |
|   | 5004        | Kiara Malhotra | F      | 29   | 2015-07-05 | NULL                           | 70000.00 | 8             | 3           |
|   | 5005        | Anvi Chaudhary | F      | 25   | 2015-09-11 | Business Development Executive | 55000.00 | 11            | 1           |
|   | 5006        | Dhruv Shetty   | M      | 28   | 2015-11-20 | UI Developer                   | 65000.00 | 8             | 2           |
| * | NULL        | NULL           | NULL   | NULL | NULL       | NULL                           | NULL     | NULL          | NULL        |

- Find the employee whose designation is missing and fill it with "Data Scientist".

## SCREENSHOT:

```

110
111 • select * from employees where designation is null;
112
113 • update employees set designation = "Data Scientist" where designation is null;
114
115 • select * from employees where designation = "Data Scientist";

```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: IA

|   | employee_id | employee_name  | gender | age  | hire_date  | designation    | salary    | department_id | location_id |
|---|-------------|----------------|--------|------|------------|----------------|-----------|---------------|-------------|
| ▶ | 5004        | Kiara Malhotra | F      | 29   | 2015-07-05 | Data Scientist | 70000.00  | 8             | 3           |
|   | 5025        | Avni Iyengar   | F      | 31   | 2019-01-20 | Data Scientist | 100000.00 | 3             | 4           |
| * | NULL        | NULL           | NULL   | NULL | NULL       | NULL           | NULL      | NULL          | NULL        |

## SORTING AND GROUPING DATA

### 1. ORDER BY:

- Find employees sorted by department ID in ascending order and salary in descending order.

## SCREENSHOT:



```

120
121                                     /* Sorting and Grouping Data */
122     /* order by */
123
124 • select * from employees order by department_id, salary desc;
125

```

| Result Grid |             |                |        |     | Filter Rows: |                               | Edit:     |               | Export/Import: |  | Wrap Cell Content: |  |
|-------------|-------------|----------------|--------|-----|--------------|-------------------------------|-----------|---------------|----------------|--|--------------------|--|
|             | employee_id | employee_name  | gender | age | hire_date    | designation                   | salary    | department_id | location_id    |  |                    |  |
| ▶           | 5020        | Aarav Nair     | M      | 32  | 2018-03-05   | Software Engineer             | 90000.00  | 1             | 1              |  |                    |  |
|             | 5007        | Anushka Singh  | F      | 32  | 2016-01-15   | Marketing Manager             | 90000.00  | 2             | 3              |  |                    |  |
|             | 5023        | Anika Paul     | F      | 30  | 2018-09-20   | Public Relations Specialist   | 70000.00  | 2             | 2              |  |                    |  |
|             | 5011        | Saanvi Patel   | F      | 28  | 2016-09-20   | Marketing Analyst             | 60000.00  | 2             | 1              |  |                    |  |
|             | 5025        | Avni Iyengar   | F      | 31  | 2019-01-20   | Data Scientist                | 100000.00 | 3             | 4              |  |                    |  |
|             | 5026        | Vivaan Singh   | M      | 29  | 2019-03-10   | Business Analyst              | 75000.00  | 3             | 2              |  |                    |  |
|             | 5001        | Vihaan Singh   | M      | 27  | 2015-01-20   | Data Analyst                  | 60000.00  | 3             | 4              |  |                    |  |
|             | 5016        | Zoya Khan      | F      | 31  | 2017-07-05   | Legal Counsel                 | 100000.00 | 4             | 4              |  |                    |  |
|             | 5022        | Aadhya Iyengar | F      | 28  | 2018-07-10   | HR Specialist                 | 60000.00  | 4             | 4              |  |                    |  |
|             | 5019        | Ishika Patel   | F      | 29  | 2018-01-15   | Talent Acquisition Specialist | 55000.00  | 4             | 4              |  |                    |  |
|             | 5024        | Aryan Shetty   | M      | 27  | 2018-11-05   | Product Manager               | 95000.00  | 5             | 1              |  |                    |  |
|             | 5030        | Sara Iyer      | F      | 28  | 2019-11-20   | Project Manager               | 90000.00  | 5             | 1              |  |                    |  |
|             | 5027        | Ananya Paul    | F      | 32  | 2019-05-20   | Content Writer                | 60000.00  | 6             | 3              |  |                    |  |
|             | 5028        | Anaya Kapoor   | F      | 26  | 2019-07-05   | Event Coordinator             | 60000.00  | 6             | 1              |  |                    |  |
|             | 5015        | Ishaan Kumar   | M      | 27  | 2017-05-20   | Financial Analyst             | 85000.00  | 7             | 1              |  |                    |  |
|             | 5021        | Advik Kapoor   | M      | 26  | 2018-05-20   | Finance Analyst               | 85000.00  | 7             | 3              |  |                    |  |
|             | 5004        | Kiara Malhotra | F      | 29  | 2015-07-05   | Data Scientist                | 70000.00  | 8             | 3              |  |                    |  |
|             | 5008        | Diya Jha       | F      | 27  | 2016-03-05   | Graphic Designer              | 70000.00  | 8             | 4              |  |                    |  |

## 2. LIMIT:

- Display the first 5 employees hired in the year 2018.

### SCREENSHOT:

126

/\* limit \*/

127

128 •

select \* from employees where hire\_date < '2018-01-01' limit 5;

Result Grid

Filter Rows:

Edit:

Export/Import:

Wrap Cell Content:

Fetch rows:

| employee_id | employee_name  | gender | age  | hire_date  | designation                    | salary   | department_id | location_id |
|-------------|----------------|--------|------|------------|--------------------------------|----------|---------------|-------------|
| 5001        | Vihaan Singh   | M      | 27   | 2015-01-20 | Data Analyst                   | 60000.00 | 3             | 4           |
| 5002        | Reyansh Singh  | M      | 31   | 2015-03-10 | Network Engineer               | 80000.00 | 12            | 1           |
| 5003        | Aaradhya Iyer  | F      | 26   | 2015-05-20 | Customer Support Executive     | 45000.00 | 10            | 2           |
| 5004        | Kiara Malhotra | F      | 29   | 2015-07-05 | Data Scientist                 | 70000.00 | 8             | 3           |
| 5005        | Anvi Chaudhary | F      | 25   | 2015-09-11 | Business Development Executive | 55000.00 | 11            | 1           |
| NULL        | NULL           | NULL   | NULL | NULL       | NULL                           | NULL     | NULL          | NULL        |

## 3. AGGREGATE FUNCTIONS:

- Calculate the sum of all salaries in the Finance department.

### SCREENSHOT:

```

129
130      /* AGGREGATE FUNCTIONS  sum & min */
131
132 •      select sum(salary) as total_finace_salary
133      from employees
134      where department_id="7";

```

| Result Grid |                     | Filter Rows: | Export: | Wrap Cell Content: |
|-------------|---------------------|--------------|---------|--------------------|
|             | total_finace_salary |              |         |                    |
| ▶           | 170000.00           |              |         |                    |

- Find the minimum age among all employees.

### SCREENSHOT:

```

136 •      select min(age) as min_age
137      from employees;

```

| Result Grid |         | Filter Rows: | Export: | Wrap Cell Content: |
|-------------|---------|--------------|---------|--------------------|
|             | min_age |              |         |                    |
| ▶           | 25      |              |         |                    |

## 4. GROUP BY:

- List the maximum salary for each location.

### SCREENSHOT:

```

139      /* GROUP BY */
140
141 •    select location_id,
142      max(salary) as max_salary
143      from employees
144      group by location_id;

```

| Result Grid |             |            | Filter Rows: | Export: |
|-------------|-------------|------------|--------------|---------|
|             | location_id | max_salary |              |         |
| ▶           | 1           | 95000.00   |              |         |
|             | 2           | 95000.00   |              |         |
|             | 3           | 90000.00   |              |         |
|             | 4           | 100000.00  |              |         |

- Calculate the average salary for each designation containing the word 'Analyst'.

## SCREENSHOT:

```

146      /* Average*/
147
148 •    select designation,
149      avg(salary) as avg_salary
150      from employees
151      where designation like '%Analyst%'
152      group by designation;

```

| Result Grid |                           |              | Filter Rows: | Export: | Wrap Cell Content: |
|-------------|---------------------------|--------------|--------------|---------|--------------------|
|             | designation               | avg_salary   |              |         |                    |
| ▶           | Data Analyst              | 60000.000000 |              |         |                    |
|             | Marketing Analyst         | 60000.000000 |              |         |                    |
|             | Supply Chain Analyst      | 60000.000000 |              |         |                    |
|             | Financial Analyst         | 85000.000000 |              |         |                    |
|             | Finance Analyst           | 85000.000000 |              |         |                    |
|             | Business Analyst          | 75000.000000 |              |         |                    |
|             | Quality Assurance Analyst | 80000.000000 |              |         |                    |

## 5. HAVING:

- Find departments with less than 3 employees.

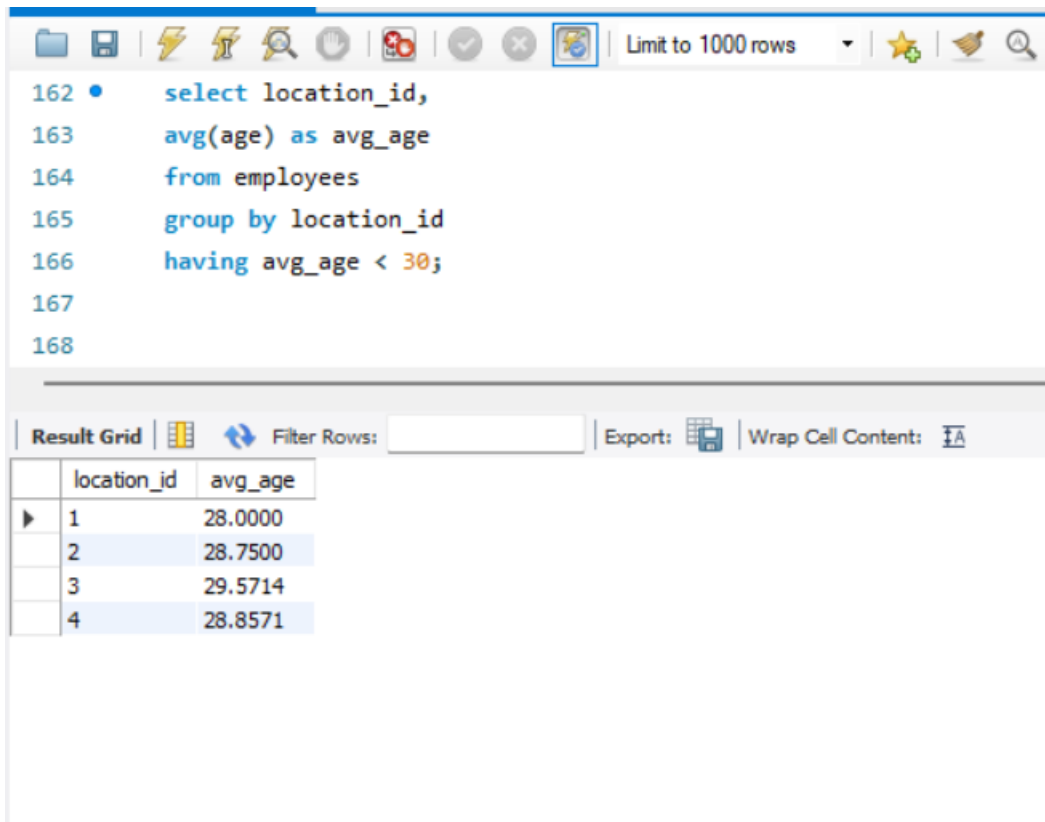
### SCREENSHOT:

```
154      /* Having */
155
156 •   select department_id,
157      count(*) as employee_count
158   from employees
159   group by department_id
160   having COUNT(*) < 3;
```

| Result Grid   | Filter Rows:   | Export: | Wrap Cell Content: |
|---------------|----------------|---------|--------------------|
| department_id | employee_count |         |                    |
| 1             | 1              |         |                    |
| 5             | 2              |         |                    |
| 6             | 2              |         |                    |
| 7             | 2              |         |                    |
| 9             | 1              |         |                    |
| 11            | 2              |         |                    |
| 13            | 1              |         |                    |

- Find locations with female employees whose average age is below 30.

### SCREENSHOT:



The screenshot shows a SQL IDE interface. The top toolbar includes icons for file operations, execution, and search, along with a 'Limit to 1000 rows' dropdown. The SQL editor contains the following query:

```
162 • select location_id,  
163      avg(age) as avg_age  
164      from employees  
165      group by location_id  
166      having avg_age < 30;  
167  
168
```

Below the editor is the 'Result Grid' section, which includes a 'Filter Rows' input field, an 'Export' button, and a 'Wrap Cell Content' checkbox. The results are displayed in a table with two columns: 'location\_id' and 'avg\_age'.

|   | location_id | avg_age |
|---|-------------|---------|
| ▶ | 1           | 28.0000 |
|   | 2           | 28.7500 |
|   | 3           | 29.5714 |
|   | 4           | 28.8571 |

## JOINS

### 1. INNER JOIN:

- List employee names, their designations, and department names where employees are assigned to a department.

**SCREENSHOT:**

```

170      /* INNER JOIN */
171 •    select e.employee_name,e.designation,d.department_name
172      from employees as e
173      inner join departments as d on e.department_id=d.department_id;
174

```

| Result Grid     Filter Rows: <input type="text"/>   Export:    Wrap Cell Content:  |                |                                |                      |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|--------------------------------|----------------------|
|                                                                                                                                                                                                                                                                                                                                                                                                                          | employee_name  | designation                    | department_name      |
| •                                                                                                                                                                                                                                                                                                                                                                                                                        | Anvi Chaudhary | Business Development Executive | Business Development |
|                                                                                                                                                                                                                                                                                                                                                                                                                          | Kiaan Desai    | Sales Executive                | Business Development |
|                                                                                                                                                                                                                                                                                                                                                                                                                          | Ananya Paul    | Content Writer                 | Content Creation     |
|                                                                                                                                                                                                                                                                                                                                                                                                                          | Anaya Kapoor   | Event Coordinator              | Content Creation     |
|                                                                                                                                                                                                                                                                                                                                                                                                                          | Aaradhya Iyer  | Customer Support Executive     | Customer Support     |
|                                                                                                                                                                                                                                                                                                                                                                                                                          | Arnav Rao      | Customer Success Manager       | Customer Support     |
|                                                                                                                                                                                                                                                                                                                                                                                                                          | Vihaan Mohan   | Supply Chain Analyst           | Customer Support     |
|                                                                                                                                                                                                                                                                                                                                                                                                                          | Vihaan Singh   | Data Analyst                   | Data Science         |
|                                                                                                                                                                                                                                                                                                                                                                                                                          | Avni Iyengar   | Data Scientist                 | Data Science         |
|                                                                                                                                                                                                                                                                                                                                                                                                                          | Vivaan Singh   | Business Analyst               | Data Science         |
|                                                                                                                                                                                                                                                                                                                                                                                                                          | Kiara Malhotra | Data Scientist                 | Design               |
|                                                                                                                                                                                                                                                                                                                                                                                                                          | Dhruv Shetty   | UI Developer                   | Design               |
|                                                                                                                                                                                                                                                                                                                                                                                                                          | Diya Jha       | Graphic Designer               | Design               |
|                                                                                                                                                                                                                                                                                                                                                                                                                          | Ishaan Kumar   | Financial Analyst              | Finance              |
|                                                                                                                                                                                                                                                                                                                                                                                                                          | Advik Kapoor   | Finance Analyst                | Finance              |
|                                                                                                                                                                                                                                                                                                                                                                                                                          | Zoya Khan      | Legal Counsel                  | Human Resources      |
|                                                                                                                                                                                                                                                                                                                                                                                                                          | Ishika Patel   | Talent Acquisition Specialist  | Human Resources      |
|                                                                                                                                                                                                                                                                                                                                                                                                                          | Aadhya Iyengar | HR Specialist                  | Human Resources      |

## 2. LEFT JOIN:

- List all departments along with the total number of employees in each department, including departments with no employees.

**SCREENSHOT:**

```

175      /* LEFT JOIN */
176 •    select d.department_name,
177      count(e.employee_id) as total_employees
178      from departments d
179      left join employees e
180      on d.department_id = e.department_id
181      group by d.department_name;

```

| Result Grid            |                 |            |
|------------------------|-----------------|------------|
|                        | Filter Rows:    | Export: Wn |
| department_name        | total_employees |            |
| Business Development   | 2               |            |
| Content Creation       | 2               |            |
| Customer Support       | 3               |            |
| Data Science           | 3               |            |
| Design                 | 3               |            |
| Finance                | 2               |            |
| Human Resources        | 3               |            |
| IT                     | 4               |            |
| Marketing              | 3               |            |
| Operations             | 1               |            |
| Product Management     | 2               |            |
| Research and Develo... | 1               |            |
| Software Development   | 1               |            |

### 3. RIGHT JOIN:

- Display all locations along with the names of employees assigned to each location. If no employees are assigned to a location, display NULL for employee name.

**SCREENSHOT:**

Employee Database S2\*

Limit to 1000 rows

```

182
183  /* RIGHT JOIN */
184 • select l.location_name,e.employee_name from employees e
185 right join location l
186 on e.location_id = l.location_id;
187
188

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: [\[A\]](#)

|   | location_name | employee_name  |
|---|---------------|----------------|
| ▶ | Bangalore     | Aaradhya Iyer  |
|   | Bangalore     | Dhruv Shetty   |
|   | Bangalore     | Myra Verma     |
|   | Bangalore     | Vihaan Mohan   |
|   | Bangalore     | Kabir Nair     |
|   | Bangalore     | Anika Paul     |
|   | Bangalore     | Vivaan Singh   |
|   | Bangalore     | Arjun Kumar    |
|   | Chennai       | Reyansh Singh  |
|   | Chennai       | Anvi Chaudhary |
|   | Chennai       | Saanvi Patel   |
|   | Chennai       | Ishaan Kumar   |
|   | Chennai       | Aarav Nair     |
|   | Chennai       | Aryan Shetty   |
|   | Chennai       | Anaya Kapoor   |
|   | Chennai       | Sara Iyer      |
|   | Hyderabad     | Kiara Malhotra |
|   | Hyderabad     | Anushka Singh  |

#### 4. CROSS JOIN

- Show all possible combinations of departments and locations.

**SCREENSHOT:**



```

191      /* CROSS JOIN */
192
193 •   select d.department_name,l.location_name from departments d
194      cross join location l;

```

| Result Grid |                      | Filter Rows:  | Export: | Wrap Cell Content: |
|-------------|----------------------|---------------|---------|--------------------|
|             | department_name      | location_name |         |                    |
| ▶           | Business Development | Pune          |         |                    |
|             | Business Development | Hyderabad     |         |                    |
|             | Business Development | Chennai       |         |                    |
|             | Business Development | Bangalore     |         |                    |
|             | Content Creation     | Pune          |         |                    |
|             | Content Creation     | Hyderabad     |         |                    |
|             | Content Creation     | Chennai       |         |                    |
|             | Content Creation     | Bangalore     |         |                    |
|             | Customer Support     | Pune          |         |                    |
|             | Customer Support     | Hyderabad     |         |                    |
|             | Customer Support     | Chennai       |         |                    |
|             | Customer Support     | Bangalore     |         |                    |
|             | Data Science         | Pune          |         |                    |
|             | Data Science         | Hyderabad     |         |                    |
|             | Data Science         | Chennai       |         |                    |
|             | Data Science         | Bangalore     |         |                    |
|             | Design               | Pune          |         |                    |
|             | Design               | Hyderabad     |         |                    |

## 5. SELF JOIN:

- Show pairs of employees working in the same department, excluding self-pairs.

**SCREENSHOT:**

```

199 • select e1.employee_name as employee1,
200         e2.employee_name as employee2,
201         e1.department_id
202 from employees e1
203 join employees e2
204 on e1.department_id = e2.department_id
205 and e1.employee_id < e2.employee_id;

```

Result Grid |  Filter Rows:  | Export:  | Wrap Cell Content: 

|   | employee1      | employee2    | department_id |
|---|----------------|--------------|---------------|
| ▶ | Vihaan Singh   | Avni Iyengar | 3             |
|   | Vihaan Singh   | Vivaan Singh | 3             |
|   | Reyansh Singh  | Atharv Yadav | 12            |
|   | Reyansh Singh  | Kabir Nair   | 12            |
|   | Reyansh Singh  | Arjun Kumar  | 12            |
|   | Aaradhya Iyer  | Arnav Rao    | 10            |
|   | Aaradhya Iyer  | Vihaan Mohan | 10            |
|   | Kiara Malhotra | Dhruv Shetty | 8             |
|   | Kiara Malhotra | Diya Jha     | 8             |
|   | Anvi Chaudhary | Kiaan Desai  | 11            |
|   | Dhruv Shetty   | Diya Jha     | 8             |
|   | Anushka Singh  | Saanvi Patel | 2             |
|   | Anushka Singh  | Anika Paul   | 2             |
|   | Atharv Yadav   | Kabir Nair   | 12            |
|   | Atharv Yadav   | Arjun Kumar  | 12            |
|   | Saanvi Patel   | Anika Paul   | 2             |
|   | Arnav Rao      | Vihaan Mohan | 10            |
|   | Ishaan Kumar   | Advik Kapoor | 7             |

## WINDOWS FUNCTION

- Write a window function query to rank employees by salary using rank().

**SCREENSHOT:**

```

210 • select employee_id,employee_name,salary,
211        rank() over (order by salary desc) as salary_rank
212 from employees;

```

Result Grid |  Filter Rows:  | Export:  | Wrap Cell Content: 

| employee_id | employee_name  | salary    | salary_rank |
|-------------|----------------|-----------|-------------|
| 5016        | Zoya Khan      | 100000.00 | 1           |
| 5025        | Avni Iyengar   | 100000.00 | 1           |
| 5012        | Myra Verma     | 95000.00  | 3           |
| 5024        | Aryan Shetty   | 95000.00  | 3           |
| 5007        | Anushka Singh  | 90000.00  | 5           |
| 5020        | Aarav Nair     | 90000.00  | 5           |
| 5030        | Sara Iyer      | 90000.00  | 5           |
| 5015        | Ishaan Kumar   | 85000.00  | 8           |
| 5021        | Advik Kapoor   | 85000.00  | 8           |
| 5002        | Reyansh Singh  | 80000.00  | 10          |
| 5010        | Atharv Yadav   | 80000.00  | 10          |
| 5017        | Kabir Nair     | 80000.00  | 10          |
| 5029        | Arjun Kumar    | 80000.00  | 10          |
| 5013        | Arnav Rao      | 75000.00  | 14          |
| 5018        | Ishan Mishra   | 75000.00  | 14          |
| 5026        | Vivaan Singh   | 75000.00  | 14          |
| 5004        | Kiara Malhotra | 70000.00  | 17          |
| 5008        | Diya Jha       | 70000.00  | 17          |

- Write a window function query to rank employees by salary within each department using DENSE RANK()

**SCREENSHOT:**

```

214      /* DENSE_RANK */
215
216 •    select employee_id,employee_name,department_id,salary,
217       dense_rank() over (partition by department_id order by salary desc) as salary_rank
218       from employees;
219

```

| Result Grid    Filter Rows: <input type="text"/>   Export:    Wrap Cell Content:  |             |                |               |           |             |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|----------------|---------------|-----------|-------------|
|                                                                                                                                                                                                                                                                                                                                        | employee_id | employee_name  | department_id | salary    | salary_rank |
| ▶                                                                                                                                                                                                                                                                                                                                      | 5020        | Aarav Nair     | 1             | 90000.00  | 1           |
|                                                                                                                                                                                                                                                                                                                                        | 5007        | Anushka Singh  | 2             | 90000.00  | 1           |
|                                                                                                                                                                                                                                                                                                                                        | 5023        | Anika Paul     | 2             | 70000.00  | 2           |
|                                                                                                                                                                                                                                                                                                                                        | 5011        | Saanvi Patel   | 2             | 60000.00  | 3           |
|                                                                                                                                                                                                                                                                                                                                        | 5025        | Avni Iyengar   | 3             | 100000.00 | 1           |
|                                                                                                                                                                                                                                                                                                                                        | 5026        | Vivaan Singh   | 3             | 75000.00  | 2           |
|                                                                                                                                                                                                                                                                                                                                        | 5001        | Vihaan Singh   | 3             | 60000.00  | 3           |
|                                                                                                                                                                                                                                                                                                                                        | 5016        | Zoya Khan      | 4             | 100000.00 | 1           |
|                                                                                                                                                                                                                                                                                                                                        | 5022        | Aadhya Iyengar | 4             | 60000.00  | 2           |
|                                                                                                                                                                                                                                                                                                                                        | 5019        | Ishika Patel   | 4             | 55000.00  | 3           |
|                                                                                                                                                                                                                                                                                                                                        | 5024        | Aryan Shetty   | 5             | 95000.00  | 1           |
|                                                                                                                                                                                                                                                                                                                                        | 5030        | Sara Iyer      | 5             | 90000.00  | 2           |
|                                                                                                                                                                                                                                                                                                                                        | 5027        | Ananya Paul    | 6             | 60000.00  | 1           |
|                                                                                                                                                                                                                                                                                                                                        | 5028        | Anaya Kapoor   | 6             | 60000.00  | 1           |
|                                                                                                                                                                                                                                                                                                                                        | 5015        | Ishaan Kumar   | 7             | 85000.00  | 1           |
|                                                                                                                                                                                                                                                                                                                                        | 5021        | Advik Kapoor   | 7             | 85000.00  | 1           |
|                                                                                                                                                                                                                                                                                                                                        | 5004        | Kiara Malhotra | 8             | 70000.00  | 1           |

- Write a window function query, Running total salary by department

**SCREENSHOT:**

```

220 • select employee_id,employee_name,department_id,salary,
221 SUM(salary) over (partition by department_id order by employee_id) as running_total_salary
222 from employees;

```

| Result Grid                                                                                                                                                                                    |                |               |           |                      |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|---------------|-----------|----------------------|
| Filter Rows: <input type="text"/>                                                                                                                                                              |                |               |           |                      |
| Export:  Wrap Cell Content:  |                |               |           |                      |
| employee_id                                                                                                                                                                                    | employee_name  | department_id | salary    | running_total_salary |
| 5020                                                                                                                                                                                           | Aarav Nair     | 1             | 90000.00  | 90000.00             |
| 5007                                                                                                                                                                                           | Anushka Singh  | 2             | 90000.00  | 90000.00             |
| 5011                                                                                                                                                                                           | Saanvi Patel   | 2             | 60000.00  | 150000.00            |
| 5023                                                                                                                                                                                           | Anika Paul     | 2             | 70000.00  | 220000.00            |
| 5001                                                                                                                                                                                           | Vihaan Singh   | 3             | 60000.00  | 60000.00             |
| 5025                                                                                                                                                                                           | Avni Iyengar   | 3             | 100000.00 | 160000.00            |
| 5026                                                                                                                                                                                           | Vivaan Singh   | 3             | 75000.00  | 235000.00            |
| 5016                                                                                                                                                                                           | Zoya Khan      | 4             | 100000.00 | 100000.00            |
| 5019                                                                                                                                                                                           | Ishika Patel   | 4             | 55000.00  | 155000.00            |
| 5022                                                                                                                                                                                           | Aadhya Iyengar | 4             | 60000.00  | 215000.00            |
| 5024                                                                                                                                                                                           | Aryan Shetty   | 5             | 95000.00  | 95000.00             |
| 5030                                                                                                                                                                                           | Sara Iyer      | 5             | 90000.00  | 185000.00            |
| 5027                                                                                                                                                                                           | Ananya Paul    | 6             | 60000.00  | 60000.00             |
| 5028                                                                                                                                                                                           | Anaya Kapoor   | 6             | 60000.00  | 120000.00            |
| 5015                                                                                                                                                                                           | Ishaan Kumar   | 7             | 85000.00  | 85000.00             |
| 5021                                                                                                                                                                                           | Advik Kapoor   | 7             | 85000.00  | 170000.00            |
| 5004                                                                                                                                                                                           | Kiara Malhotra | 8             | 70000.00  | 70000.00             |

## 8. Key Outcomes:

- Strong understanding of SQL fundamentals
- Hands-on experience with database design
- Practical knowledge of joins and relationships
- Ability to write optimized queries
- Exposure to advanced SQL concepts like window functions

## 9. Conclusion

This project successfully demonstrates the implementation of an Employee Management System using SQL. It covers both basic and advanced SQL operations, helping build a solid foundation in database management and data analysis.

The knowledge gained from this project can be applied to real-world data handling, reporting, and analytics tasks.